

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0405

Littlefork 138 kV
August 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0405. The Interconnection Customer proposes a 252.1 MW wind facility interconnecting at the Littlefork 138 kV substation in the Virginia area. The request was received on November 25, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 194 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,498,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Van Tyle - Riverton 230 138 KV	Riverton 230 - Electric Avenue Facility 138 KV L	128.1	233	6.0
Coyote Crk - Van Tyle 138 KV	Riverton 230 - Electric Avenue Facility 138 KV line	124.3	219	7.7
Indeck West Enfield Energy Ctr - Acadiana Mall 138 KV Line	Riverton 230 - Electric Avenue Facility 138 KV	118.7	226	6.3
Riverton 230 - Remington Ct 138 KV	Tuckasegee Tie - Merced 138 KV Line	111.5	499	10.0
Coyote Crk - Hogan Rd 138 KV	Tuckasegee Tie - Electric Avenue Facility 138 KV	109.9	208	8.9
Oak Green - Spring Gap Tap 138 KV	Electric Avenue Facility - Riverton 230 138 KV	107.2	233	7.3
Lewisburg - South Jordan 138 KV	Indeck West Enfield Energy Center - Middletown Junction 138 KV Line	104.8	285	5.2
Cholame - Van Tyle 138 KV	Riverton 230 Substation - Remngton Ct 138 KV	100.6	190	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Van Tyle - Riverton 230 138 KV	Rebuild line with higher-capacity conductor	\$3,105,000	22% (\$700,000)
Coyote Crk - Van Tyle 138 KV	Rebuild line with higher-capacity conductor	\$15,270,000	21% (\$3,184,000)
Indeck West Enfield Energy Ctr - Acadiana Mall 138 KV Line	Rebuild line with higher-capacity conductor	\$4,582,000	9% (\$414,000)
Riverton 230 - Remington Ct 138 KV Line	Rebuild line with higher-capacity conductor	\$33,797,000	8% (\$2,591,000)
Coyote Crk - Hogan Rd 138 KV	Reconductor line	\$3,870,000	11% (\$442,000)
Oak Green - Spring Gap Tap 138 KV	Reconductor line	\$2,353,000	10% (\$244,000)
Lewisburg - South Jordan 138 KV	Reconductor line	\$2,224,000	11% (\$248,000)
Van Tyle Substation - Cholame 138 KV	Reconductor line	\$7,137,000	10% (\$675,000)
	Total allocated to project		\$8,498,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.